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324614 (24)

BE (6th Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical

MICROPROCESSORS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Part (a) of each question is compulsory.

(ii) Attempt any two parts from (b), (c) and (d) in each question.

(iii) Make suitable assumptions if required.

Q. 1. (a) Write short note on evolution of microprocessors. 2

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(b) Draw the pin diagram of 8085. Group various signals of 8085 and explain them in brief. 7

(c) Explain following instructions of 8085 (in brief) : 7

(i) LDAX R_P (ii) MVI M, Data

(iii) ADC M (iv) DAD R_p

(d) Write a program to perform 16 bit addition of following data :

1 2 3 4 H

4 3 2 1 H

Store the result in H and L registers. Also draw the flow diagram. 7

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Q. 2. (a) What is the power requirement of DRAM ? 2

(b) What are semiconductor memories ? Classify them and explain in brief with the help of suitable diagrams. 7

(b) Draw a diagram to indicate various signals on HM6264 static RAM chip. Also draw the real cycle for HM6264. 7

(d) Answer following : 7

(i) Soft errors in ROMs

(ii) Shadow RAMs.

Q. 3. (a) What do you understand by "masking the interrupts" ? Explain. 2

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(b) Design a decoder circuit with the help of a 74LS138 decoder using following address map :

Address		Device to be selected
From	To	
0000	1FF H	EPROM
2000	3FF H	RAM # 0
4000	5FF H	RAM # 1
6000	7FF H	RAM # 2
8000	9FF H	RAM # 03
0A000	0BFF H	RAM # 04
0C000	0DFF H	RAM # 05
0E000	0E000 H	I/o # 1
0E001	0E001 H	I/o # 2

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Note : Use Hitahci 2764 as EPROM and Hitachi 6264 as static RAM. 7

(c) Explain following : 7

(i) Address Space Partitioning

(ii) Differentiate synchronous and asynchronous data transfer.

(d) What is DMA ? Draw the flow chart and explain in detail. 7

Q. 4. (a) What is the task performed by AD557 chip ? 2

(b) What is a port ? Explain the working of 8 Bit 8212 port with the help of a block diagram. 7

(c) Draw a diagram to indicate various signals of 8259 PIC chip. Explain in brief. How 8259 chips can be cascaded ? 7

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(d) Explain various signals of 8257 DMA controller with the help of suitable diagram. Also draw the state transition diagram for 8257. 7

Q. 5. (a) What is MDS ? Why it is required ? 2

(b) What is an "operating-system" ? What are its types ? Explain in detail with the help of diagrams. 7

(c) Explain following : 7

(i) Macro Assembler

(ii) Tasks performed by "system-monitor".

(d) What are the hardware aids for a microprocessor ? Explain with the help of suitable diagrams. 7

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